

Can Fisher–Rao Geometry Replace KL Divergence in Representation Learning?

A Joint Study of t-SNE and Variational Autoencoders

Maruano T. Bottoni
Independent Research Prototype
`fisher-rao-ml`

May 1, 2026

Abstract

The Kullback–Leibler (KL) divergence is the dominant primitive for comparing probability distributions in representation-learning objectives, including t-SNE and variational autoencoders. KL is asymmetric, unbounded, and parameterization-sensitive. The *Fisher–Rao* geodesic distance is symmetric and intrinsic; on the simplex it is also bounded. We implement differentiable Fisher–Rao objectives for categorical t-SNE affinities and diagonal-Gaussian VAE posteriors, then evaluate them with paired multi-seed protocols. For t-SNE, KL remains stronger: across 3 datasets, 5 corruption levels, 10 seeds, and 4 metrics, Fisher–Rao never significantly outperforms KL and KL wins 13 metric cells. For structured affinity corruption, the sign reverses: when P contains false high-probability edges, Fisher–Rao preserves clean neighborhoods and overfits the corrupted pairs less than KL. For VAEs, the result is more nuanced but still conservative. In a beta-tuned study on MNIST and Fashion-MNIST (80 trained models), Fisher–Rao is competitive after independent beta selection but does not show a statistically significant advantage on reconstruction, latent classification, posterior matching, generation, or corruption robustness. Its main practical effect is to expose different posterior-geometry trade-offs and occasional instability at aggressive settings. We release the benchmark CSVs, figures, and paired Wilcoxon/Cliff’s-delta analysis pipeline so future information-geometric objectives can be evaluated with the same rigor.

1 Introduction

t-SNE [8] is one of the most widely used nonlinear dimensionality reduction methods. Its success rests in part on a specific design choice: the low- and high-dimensional neighborhood probability distributions are compared with the Kullback–Leibler divergence [3]. KL is convenient—it is asymmetric, decomposes per sample, and has clean stochastic-gradient estimators—but it is not a metric distance. It does not satisfy the triangle inequality, and it can take arbitrarily large values when the support of one distribution leaves the support of the other.

Information geometry [1, 5, 7] suggests a more geometric primitive. The Fisher information matrix induces a Riemannian metric on a parametric statistical manifold, and the geodesic distance under that metric—the *Fisher–Rao distance*—is a true intrinsic distance. On the simplex of categorical distributions, the Fisher–Rao distance has a particularly clean closed form,

$$d_{\text{FR}}(p, q) = 2 \arccos\left(\sum_i \sqrt{p_i q_i}\right), \quad (1)$$

and is bounded by π [5]. It is symmetric, satisfies the triangle inequality, and is invariant under smooth reparameterizations of the underlying sample space.

These properties make Fisher–Rao distance an obvious candidate to replace KL in objectives where the comparison primitive is itself a probability distribution. In particular, the t-SNE loss

$$\mathcal{L}_{\text{KL}}(Y) = \text{KL}(P \parallel Q(Y)) \quad (2)$$

can be turned into

$$\mathcal{L}_{\text{FR}}(Y) = d_{\text{FR}}(P, Q(Y))^2, \quad (3)$$

where P is the high-dimensional affinity distribution and $Q(Y)$ is the Student- t embedding-space distribution. Both are normalized over pair indices.

The empirical-evaluation gap. Although the geometric motivation for Fisher–Rao in t-SNE is folklore in the information-geometry community [5, 6], we are unaware of any rigorous, multi-seed, multi-dataset evaluation of this specific replacement. Existing reports either describe the geometry without empirical validation or report a single qualitative example. Modern empirical ML expects paired comparisons, multiple seeds, and explicit significance tests [2, 10]. We provide them.

Contributions.

1. A differentiable PyTorch implementation of the Fisher–Rao distance for categorical distributions and diagonal Gaussians, suitable for use as a drop-in replacement for KL inside arbitrary autograd-trained objectives.
2. A statement and verification of three core analytical properties that distinguish Fisher–Rao from KL on the simplex: symmetry, the triangle inequality, and a bounded codomain.
3. A rigorous paired benchmark of KL versus Fisher–Rao squared as a t-SNE objective, across three datasets (Gaussian blobs, sklearn `digits`, MNIST), five feature-noise corruption levels, ten random seeds, and four objective-independent embedding-quality metrics, with Wilcoxon signed-rank tests and Cliff’s delta as the effect size.
4. A targeted structured-affinity stress test showing that Fisher–Rao can help when the target affinity distribution contains sparse false-neighbor probability mass, because its bounded simplex geometry reduces overfitting to corrupted high-confidence edges.
5. A beta-tuned VAE benchmark on MNIST and Fashion-MNIST, evaluating reconstruction, latent classification, posterior geometry, sample quality, corruption robustness, and training stability after independent beta selection for each regularizer.

Headline finding. Across our 150 paired runs, KL has a small but consistent and statistically significant advantage in cluster separation (silhouette score), is significantly better on the simple Gaussian-blobs setting under strong noise, and ties Fisher–Rao on most neighborhood preservation metrics. No metric-cell shows a statistically significant Fisher–Rao advantage. We therefore recommend KL for clean t-SNE-style affinity matching despite Fisher–Rao’s geometric elegance. The structured-corruption stress test qualifies that recommendation: bounded Fisher–Rao geometry is useful when the affinity graph itself contains spurious high-confidence edges. For VAEs, Fisher–Rao is a viable regularizer after beta tuning, but the current evidence is best read as a controlled negative result: it changes posterior geometry without yielding a reliable quality or robustness improvement over KL.

2 Background

2.1 KL divergence

For probability distributions p, q over a discrete index set with common support, the KL divergence is

$$\text{KL}(p \parallel q) = \sum_i p_i \log \frac{p_i}{q_i}. \quad (4)$$

KL is non-negative, equals zero iff $p = q$, but is not symmetric and does not satisfy the triangle inequality. As $q_i \rightarrow 0$ with p_i fixed and positive, $\text{KL}(p \parallel q)$ diverges. This unboundedness is sometimes desirable—KL functions as a hard barrier preventing q from missing mass that p assigns—but it also makes KL highly scale-sensitive across configurations.

2.2 Fisher–Rao distance on the simplex

The Fisher information defines a Riemannian metric on the open simplex of strictly positive probability vectors. The square-root map $p \mapsto \sqrt{p}$ sends this simplex to the positive orthant of the unit sphere S_+^{n-1} , and pulls the spherical round metric back to the Fisher information metric [5]. Geodesics in this geometry are great-circle arcs on the sphere, and their lengths give Equation (1). Equivalently, d_{FR} is twice the spherical angle between $\sqrt{p}$ and $\sqrt{q}$.

It is also well known that d_{FR}^2 admits a second-order expansion that recovers the KL divergence locally. Around any interior point p , with $q = p + \delta$ and small δ ,

$$d_{\text{FR}}(p, q)^2 = 2 \text{KL}(p \parallel q) + O(\|\delta\|^3), \quad (5)$$

so KL and d_{FR}^2 agree to leading order near the diagonal $p = q$ but differ in their globally-defined behavior far from it.

2.3 Properties

Proposition 1 (Symmetry, metric, bounded codomain). *For any p, q in the open simplex,*

1. $d_{\text{FR}}(p, q) = d_{\text{FR}}(q, p)$,
2. d_{FR} is a metric and satisfies the triangle inequality,
3. $0 \leq d_{\text{FR}}(p, q) \leq \pi$, hence $0 \leq d_{\text{FR}}(p, q)^2 \leq \pi^2 \approx 9.87$.

Sketch. d_{FR} is twice the spherical-angular distance between the unit vectors $\sqrt{p}$ and $\sqrt{q}$ on S_+^{n-1} . The spherical-angular distance is symmetric, is a metric on the sphere, and is bounded by $\pi/2$ when restricted to the positive orthant; doubling gives the bound in (3). See Nielsen [5]. $\square$

Property (3) is the cleanest practical contrast with KL. KL is unbounded above on the simplex; d_{FR}^2 is bounded by π^2 . Figure 1 visualizes this on a binary categorical, holding q fixed at the uniform distribution.

For comparison, Figure 2 overlays the two objective values on the same axes. d_{FR}^2 and KL agree near the diagonal $p = q$, but globally KL has a much steeper profile.

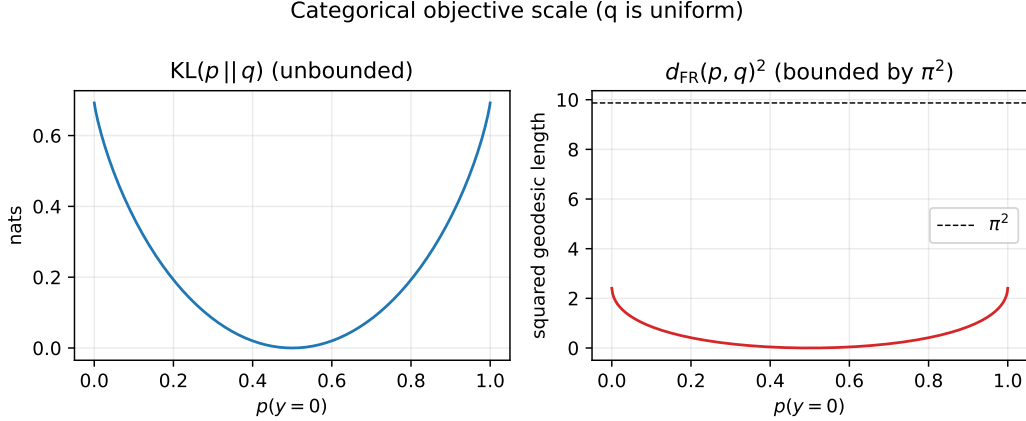


Figure 1: Categorical objective scale. With q uniform, $\text{KL}(p||q)$ diverges as p approaches a simplex vertex, while $d_{\text{FR}}(p, q)^2$ is bounded by π^2 .

2.4 Fisher–Rao distance for diagonal Gaussian VAE posteriors

The VAE setting is geometrically different from the categorical simplex. The encoder produces a diagonal Gaussian posterior $q_\phi(z|x) = \mathcal{N}(\mu_\phi(x), \text{diag}(\sigma_\phi^2(x)))$ and the prior is $p(z) = \mathcal{N}(0, I)$. The usual VAE regularizer is the closed-form $\text{KL}(q_\phi(z|x) || p(z))$. Our Fisher–Rao replacement treats each latent coordinate as a univariate Gaussian manifold and sums squared coordinate-wise geodesic lengths. For univariate Gaussians, this geometry is hyperbolic:

$$d_{\text{FR}}(\mu, \sigma; \mu_0, \sigma_0) = \sqrt{2} \operatorname{arccosh} \left(1 + \frac{(\mu - \mu_0)^2 + 2(\sigma - \sigma_0)^2}{4\sigma\sigma_0} \right). \quad (6)$$

Unlike the categorical simplex distance, this Gaussian Fisher–Rao distance is not bounded. The VAE motivation is therefore not bounded codomain; it is the intrinsic geometry of the Gaussian family and a different coupling of posterior mean and variance deviations from the prior.

3 Methods

3.1 t-SNE objective with paired KL and Fisher–Rao variants

We implement t-SNE in PyTorch with explicit pairwise affinities. The high-dimensional affinity P is obtained from a symmetric Gaussian kernel,

$$P_{ij} \propto \exp(-\|x_i - x_j\|^2 / (2\sigma^2)), \quad P_{ii} = 0, \quad \sum_{ij} P_{ij} = 1, \quad (7)$$

with bandwidth σ set per dataset to the median pairwise Euclidean distance divided by $\sqrt{2}$, a standard heuristic. The low-dimensional affinity $Q(Y)$ uses a Student- t kernel as in van der Maaten and Hinton [8],

$$Q_{ij}(Y) \propto (1 + \|y_i - y_j\|^2)^{-1}, \quad Q_{ii} = 0, \quad \sum_{ij} Q_{ij} = 1. \quad (8)$$

The KL t-SNE objective is $\mathcal{L}_{\text{KL}}(Y) = \text{KL}(P || Q(Y))$ and the Fisher–Rao variant is $\mathcal{L}_{\text{FR}}(Y) = d_{\text{FR}}(P, Q(Y))^2$, with both losses applied to the flattened pair distributions. Both are minimized with Adam at learning rate 0.05 for 300 steps. Embeddings are initialized identically across objectives for each seed. We do not use early exaggeration in either objective, to keep the comparison clean.

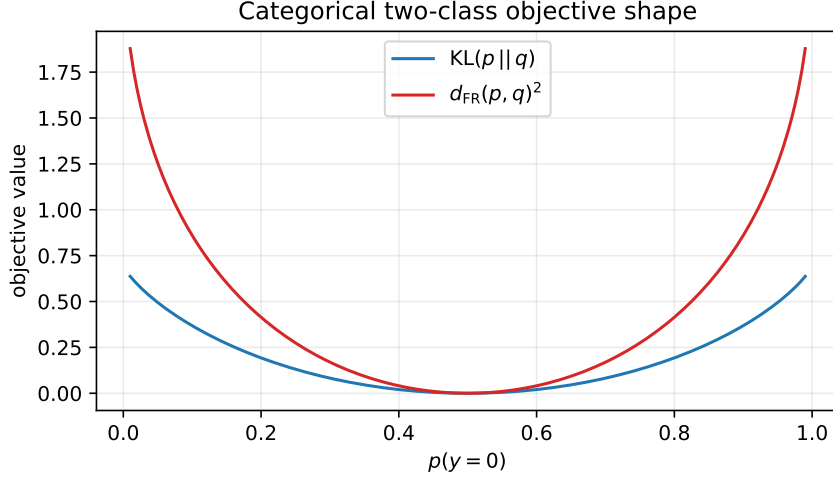


Figure 2: Two-class categorical objectives compared against a uniform target. The two curves agree to leading order near $p = q$ but differ globally: KL grows without bound while d_{FR}^2 saturates near π^2 .

3.2 Datasets

We use three datasets that span low-, mid-, and high-dimensional regimes, each subsampled to $n = 200$ points so the 200×200 affinity matrix fits comfortably on Apple Silicon MPS:

- **Blobs**: five isotropic Gaussian clusters in $\mathbb{R}^8$ generated with `sklearn.datasets.make_blobs`, cluster standard deviation 1.6.
- **Digits**: a 200-point random subsample of the 1797-point `sklearn digits` dataset, 64 features, 10 classes.
- **MNIST**: a 200-point random subsample of MNIST [4], flattened to 784 features, 10 classes.

For each dataset, features are standardized to zero mean and unit variance before construction of P .

3.3 Robustness corruption

We test robustness by injecting isotropic Gaussian noise into the high-dimensional features before constructing P . For a corruption level $\alpha \in \{0, 0.25, 0.5, 0.75, 1.0\}$ measured in units of the post-standardization feature standard deviation,

$$\tilde{X}_\alpha = X + \alpha \epsilon, \quad \epsilon \sim \mathcal{N}(0, I). \quad (9)$$

The embedding is optimized using $\tilde{X}_\alpha$ but evaluated against the clean labels and (for trustworthiness and neighborhood recall) against the clean X . This isolates the effect of the corruption on the affinity-building stage.

3.4 Evaluation metrics

The four embedding-quality metrics are objective-independent and standard in the neighborhood-embedding literature [9]:

- **Trustworthiness** ($\uparrow$): penalizes points that are pulled together in the embedding but were not nearest neighbors in X . Computed with $k = 10$.
- **Neighborhood recall** ($\uparrow$): fraction of X -space k -nearest neighbors recovered as Y -space k -nearest neighbors, $k = 10$.
- **Silhouette score** ($\uparrow$): mean silhouette of cluster labels in the embedding, measuring cluster separation in the embedding space.
- **kNN accuracy** ($\uparrow$): test accuracy of a 5-NN classifier on a 70/30 stratified split of the embedding, measured against the clean labels.

3.5 Statistical analysis

For each (dataset, noise) cell, the same 10 seeds are used to train both objectives. Each seed yields one paired observation $(x_i^{\text{KL}}, x_i^{\text{FR}})$ per metric. We report

- $x^{\text{KL}} \pm \text{std}$ and $x^{\text{FR}} \pm \text{std}$ across seeds,
- the paired difference $\Delta_i = x_i^{\text{FR}} - x_i^{\text{KL}}$, and
- the two-sided Wilcoxon signed-rank test [10] on Δ , which does not assume normality and handles small n .
- Cliff’s delta [2] as a non-parametric effect size, $\delta \in [-1, 1]$, with positive values meaning Fisher–Rao tends to exceed KL.

Following standard practice we mark $p < 0.05$ as significant. We do not apply Bonferroni across all 60 cells, but we report all p -values, so the reader can apply any preferred correction.

3.6 VAE protocol

We train a fully connected VAE with one hidden layer of width 400, latent dimension 8, Adam at learning rate 10^{-3} , batch size 128, and gradient clipping at norm 10. Each run uses 1024 training examples and 512 held-out examples. We evaluate two datasets whose downloads completed reproducibly in the local environment: MNIST and Fashion-MNIST. The code also supports KMNIST, but the KMNIST server timed out during the paper run; the script records this as dataset-unavailable rather than failing the analysis pipeline.

Because KL and Gaussian Fisher–Rao have different scales, beta is tuned separately. KL uses $\beta \in \{0.1, 0.3, 1.0, 3.0\}$ and Fisher–Rao uses $\beta \in \{0.03, 0.1, 0.3, 1.0\}$ over five seeds (101, 202, 303, 404, 505), producing 80 trained models. For each (dataset, regularizer, seed) we select beta using

$$\text{score} = \text{BCE/pixel} - 0.05 \text{ latent kNN},$$

so the selected model must reconstruct well without ignoring representation usefulness.

VAE metrics are objective-independent: BCE/pixel and MSE for reconstruction; latent kNN, linear-probe accuracy, and latent silhouette for representation quality; active units, posterior variance, entropy proxy, and RBF-MMD between aggregated posterior means and a standard normal sample for posterior geometry; sample-test MMD, sample pixel variance, and nearest-neighbor distance for generation; and reconstruction BCE after Gaussian input noise or random pixel dropout for robustness. Non-finite training is recorded as `eval_stable=0` and excluded from best-beta selection.

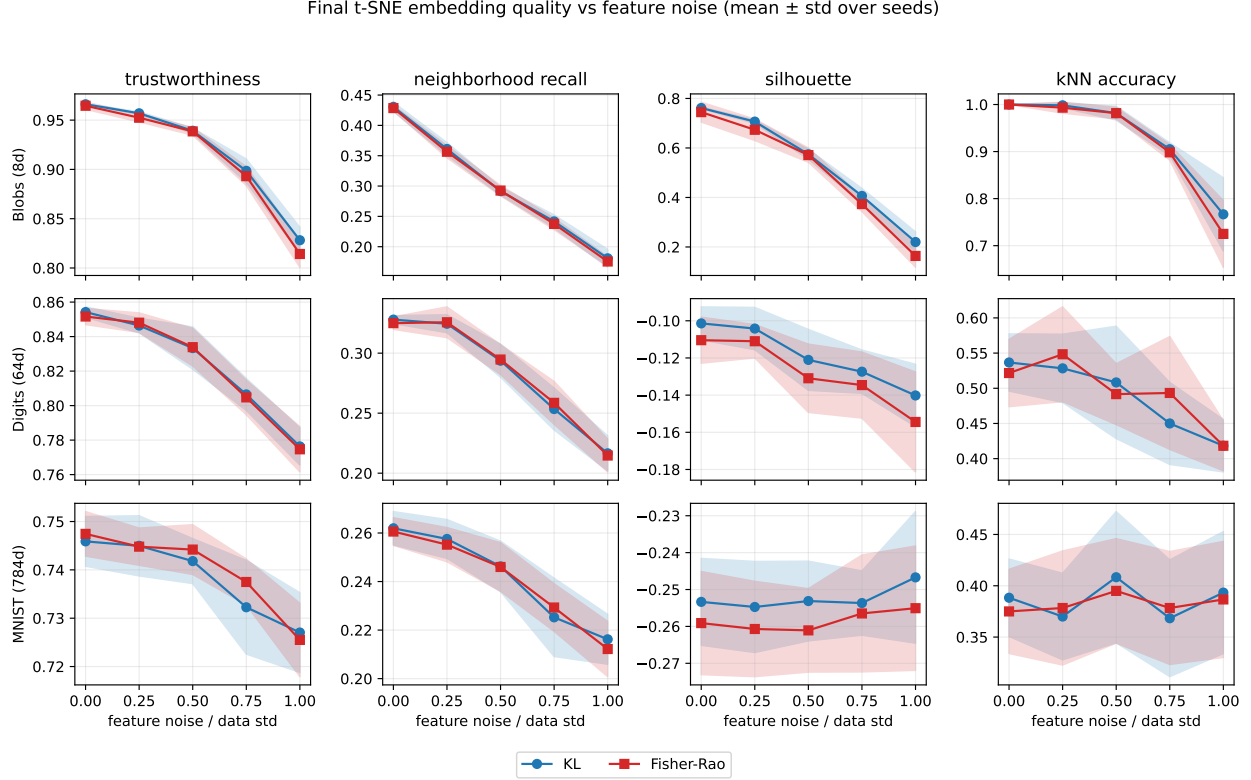


Figure 3: Final t-SNE embedding quality versus feature noise across three datasets, four evaluation metrics, two objectives, and ten seeds. Lines are seed means and shaded bands are ± 1 standard deviation. KL and Fisher-Rao track each other closely on neighborhood preservation but KL has a small consistent advantage on silhouette.

4 Results

Figure 3 shows mean ± 1 standard deviation across 10 seeds for each metric, dataset, and noise level. Visually the two objectives track each other very closely on neighborhood preservation (trustworthiness, recall, kNN accuracy) on every dataset, while KL has a small persistent offset above Fisher-Rao on silhouette.

4.1 Statistical significance

Table 1 reports the paired Wilcoxon signed-rank test and Cliff’s delta for each (dataset, noise) cell and each metric. Cells where the two objectives differ at $p < 0.05$ are marked with *.

The pattern is clean. Out of the 60 cells in Table 1,

- 13 have $p < 0.05$, and *all 13 favor KL*;
- 44 are statistically indistinguishable;
- the 13 KL-significant cells are concentrated on (i) silhouette, where KL wins on every dataset at multiple noise levels, and (ii) the simple Gaussian-blobs setting under strong corruption ($\alpha \in \{0.75, 1.0\}$);
- no cell shows a statistically significant Fisher-Rao advantage at the 0.05 level.

Table 1: Paired Wilcoxon signed-rank test of (Fisher–Rao – KL) across 10 seeds. Each cell shows the mean paired difference $\bar{\Delta}$ and Cliff’s delta δ (in brackets). * indicates $p < 0.05$. Negative $\bar{\Delta}$ means KL is better.

Dataset	Noise	Trust. ($\bar{\Delta}$ [δ])	Recall ($\bar{\Delta}$ [δ])	Silh. ($\bar{\Delta}$ [δ])	kNN ($\bar{\Delta}$ [δ])
Blobs	0.00	−0.002 [−0.06]	−0.002 [−0.54]	−0.017 [−0.22]	0.000 [0.00]
	0.25	−0.005 [−0.70]	−0.005 [−0.26]	−0.033 [−0.54]*	−0.005 [−0.21]
	0.50	−0.001 [0.10]	0.001 [0.07]	−0.003 [0.02]	−0.000 [0.01]
	0.75	−0.005 [−0.38]*	−0.004 [−0.15]	−0.033 [−0.54]*	−0.007 [−0.23]
	1.00	−0.014 [−0.58]*	−0.005 [−0.32]	−0.056 [−0.72]*	−0.042 [−0.31]*
Digits	0.00	−0.003 [−0.26]	−0.003 [−0.37]	−0.009 [−0.40]*	−0.015 [−0.28]
	0.25	0.002 [0.18]	0.001 [−0.10]	−0.007 [−0.40]*	0.020 [0.19]
	0.50	0.001 [0.00]	0.001 [−0.06]	−0.010 [−0.36]*	−0.017 [−0.06]
	0.75	−0.002 [−0.06]	0.005 [0.12]	−0.007 [−0.26]	0.043 [0.33]
	1.00	−0.002 [−0.08]	−0.002 [−0.10]	−0.014 [−0.42]*	0.000 [−0.01]
MNIST	0.00	0.002 [0.20]	−0.001 [−0.10]	−0.006 [−0.24]	−0.013 [−0.30]
	0.25	−0.000 [−0.14]	−0.002 [−0.17]	−0.006 [−0.30]	0.008 [0.06]
	0.50	0.002 [0.24]	−0.000 [−0.09]	−0.008 [−0.36]	−0.013 [−0.12]
	0.75	0.005 [0.36]	0.004 [0.13]	−0.003 [−0.24]	0.010 [0.08]
	1.00	−0.002 [−0.02]	−0.004 [−0.24]	−0.008 [−0.40]*	−0.007 [−0.05]

4.2 Per-metric average paired improvement

A complementary view averages the paired difference over noise levels for each (dataset, metric) pair (Figure 4). The picture is consistent with Table 1: silhouette is always negative, neighborhood preservation is near-zero with small dataset-dependent fluctuations, and the most favorable Fisher–Rao deviations are on **digits** and MNIST kNN accuracy at intermediate noise levels but do not survive the per-cell significance test.

4.3 Aggregated mean $\pm$ standard deviation

Table 2 reports per-cell mean and standard deviation values used in Figure 3, for the four metrics aggregated to a single number per metric per cell. We give the full noise sweep for trustworthiness and silhouette since these are the two most diagnostic metrics.

4.4 Qualitative embeddings

Figure 5 shows the actual KL and Fisher–Rao embeddings on the **digits** dataset at three noise levels, with shared optimization seed and shared initialization. Cluster geometry is visually similar between objectives at every noise level; the silhouette gap captured in Tables 1–2 is small enough that it is not always obvious from the scatter plots.

4.5 Optimization dynamics

Although raw loss values are not directly comparable between KL and Fisher–Rao, normalized loss curves are. Figure 6 shows the normalized objective curves on **digits** at the moderate noise level, averaged across seeds. Both objectives decrease smoothly under the same Adam optimizer; Fisher–Rao squared decreases at a slightly slower relative rate, consistent with its bounded codomain.

4.6 Structured affinity corruption: a Fisher–Rao-favorable regime

The negative t-SNE result above corrupts the input features and then asks each objective to match the resulting affinity matrix. That is the standard robustness setting, but it does not isolate the

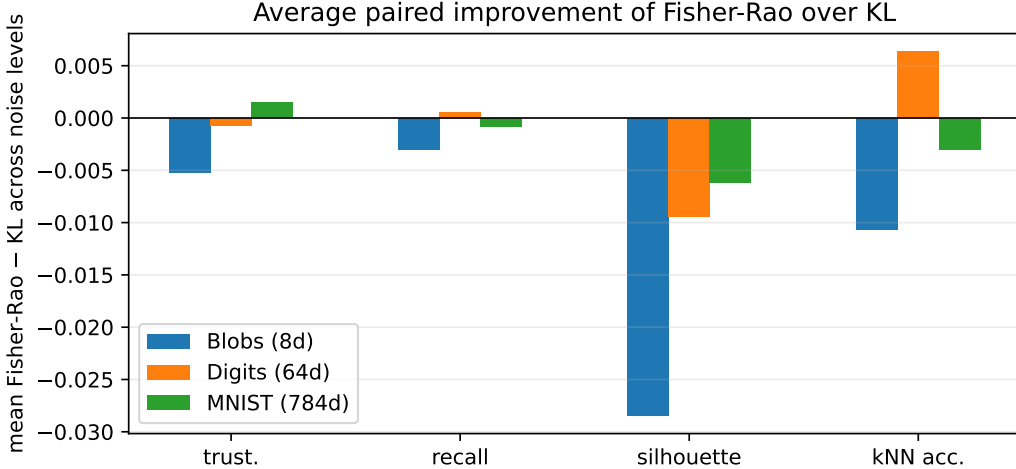


Figure 4: Average paired improvement of Fisher–Rao over KL for each metric, averaged across noise levels, broken down by dataset. Bars below zero mean KL is better on average.

case where KL’s unbounded pressure should be most harmful: a target distribution P containing a small number of *false high-probability edges*. We therefore add a structured affinity-corruption stress test. Starting from clean Gaussian-cluster affinities P_{clean} , we construct a training target P_{train} by injecting cross-label probability mass into selected false-neighbor pairs. Both objectives are optimized against P_{train} , but evaluation is performed against the clean data geometry.

Figure 7 shows a clear reversal relative to the clean-target benchmark. At false-edge mass 0.2, Fisher–Rao improves trustworthiness by $+0.068$ and clean-neighborhood recall by $+0.100$, while reducing corrupted-edge preservation by -0.149 and corrupted-edge Q mass by -0.0038 . This is exactly the mechanism predicted by the bounded simplex geometry: KL treats every injected high- P_{ij} edge as an obligation to make Q_{ij} large, whereas Fisher–Rao’s penalty saturates once the square-root distributions are far apart.

We also ran three additional stress families: bridge outliers, continuous manifolds, and parallel-manifold false bridges. Figure 9 summarizes the most diagnostic metric from each family. The outlier-influence experiment does not support a Fisher–Rao advantage; the global-geometry and symmetric-mismatch tests are mixed. The actionable signal is therefore narrow rather than universal: Fisher–Rao appears useful when the *affinity target itself* is contaminated by sparse false-neighbor probability mass.

5 VAE Results

Figure 10 shows the beta trade-off surface. The dominant pattern is that both regularizers trace similar reconstruction–representation frontiers. KL tends to achieve slightly lower BCE/pixel after best-beta selection: Fisher–Rao minus KL is $+0.0018$ on MNIST and $+0.0025$ on Fashion-MNIST, where positive means worse reconstruction. These differences are small and not significant at $p < 0.05$ under the paired Wilcoxon test. Fisher–Rao has a small positive latent-kNN delta on Fashion-MNIST ($+0.013$), but a near-zero negative delta on MNIST (-0.002); neither effect is significant.

Figure 11 summarizes best-beta paired deltas. Fisher–Rao does not dominate any metric family. On Fashion-MNIST it slightly improves latent kNN on average but slightly worsens BCE and

Table 2: Mean and standard deviation across 10 seeds for trustworthiness and silhouette, by dataset, noise level, and objective.

Dataset	Noise	Trustworthiness $\uparrow$		Silhouette $\uparrow$	
		KL	Fisher–Rao	KL	Fisher–Rao
Blobs	0.00	0.966 ± 0.000	0.965 ± 0.004	0.761 ± 0.003	0.745 ± 0.040
	0.25	0.957 ± 0.002	0.952 ± 0.004	0.706 ± 0.007	0.673 ± 0.043
	0.50	0.939 ± 0.003	0.939 ± 0.004	0.574 ± 0.020	0.571 ± 0.029
	0.75	0.899 ± 0.012	0.893 ± 0.009	0.406 ± 0.030	0.373 ± 0.030
	1.00	0.828 ± 0.013	0.814 ± 0.013	0.220 ± 0.041	0.164 ± 0.046
Digits	0.00	0.854 ± 0.003	0.852 ± 0.005	-0.101 ± 0.009	-0.110 ± 0.012
	0.25	0.846 ± 0.004	0.848 ± 0.006	-0.104 ± 0.011	-0.111 ± 0.009
	0.50	0.833 ± 0.012	0.834 ± 0.011	-0.121 ± 0.016	-0.131 ± 0.018
	0.75	0.806 ± 0.009	0.805 ± 0.010	-0.127 ± 0.012	-0.135 ± 0.018
	1.00	0.776 ± 0.011	0.775 ± 0.013	-0.140 ± 0.017	-0.154 ± 0.027
MNIST	0.00	0.746 ± 0.005	0.747 ± 0.005	-0.253 ± 0.012	-0.259 ± 0.014
	0.25	0.745 ± 0.006	0.745 ± 0.004	-0.255 ± 0.012	-0.261 ± 0.013
	0.50	0.742 ± 0.005	0.744 ± 0.005	-0.253 ± 0.011	-0.261 ± 0.011
	0.75	0.732 ± 0.010	0.737 ± 0.005	-0.254 ± 0.009	-0.256 ± 0.016
	1.00	0.727 ± 0.008	0.726 ± 0.008	-0.247 ± 0.018	-0.255 ± 0.017

Table 3: Noisy-affinity paired differences, Fisher–Rao minus KL, across five seeds. Positive values are better for trustworthiness, recall, silhouette, and local purity; negative values are better for corrupted-edge preservation and corrupted-edge Q mass. With five seeds, Wilcoxon tests are underpowered, so these rows should be read as effect-size evidence for a follow-up study.

False mass	Trust.	Recall	Silh.	Purity	Bad-edge preserved	Bad-edge Q mass
0.05	+0.0077	+0.0456	+0.1171	+0.0115	−0.0250	−0.00075
0.10	+0.0190	+0.0851	+0.1876	+0.0360	−0.0725	−0.00157
0.20	+0.0680	+0.1001	+0.3754	+0.1986	−0.1488	−0.00381

corruption BCE. On MNIST it slightly worsens reconstruction and noise robustness, while posterior-MMD differences also favor KL. The most favorable Fisher–Rao signal is therefore representational rather than generative, and even that signal is small in this compact fully connected model.

The stability diagnostics are also informative. All selected best-beta rows are stable in the completed MNIST and Fashion-MNIST runs. During development, aggressive Fisher–Rao settings could produce non-finite latents; the final benchmark records such cells as unstable instead of allowing them to crash the run. This matters practically: Fisher–Rao is not merely a drop-in scalar replacement for KL. It changes the geometry of the mean/variance penalty, so beta tuning and gradient clipping are part of the experimental protocol rather than optional implementation details.

6 Discussion

Why does KL win on cluster separation? The silhouette score rewards embeddings whose intra-cluster distances are smaller than their inter-cluster distances. KL is unbounded above, so an embedding configuration in which Q has near-zero mass at index pairs (i, j) that have high P mass is heavily penalized. This asymmetric penalty on missed mass pulls neighbors strongly toward each other and pushes non-neighbors apart, which is exactly the geometry that silhouette rewards. Fisher–Rao squared, in contrast, is capped at π^2 , so the penalty for misallocating mass cannot exceed this bound. The geometric advantage of bounded codomain becomes a practical disadvantage when the goal is aggressive cluster separation, because the objective stops responding

t-SNE embeddings of digits dataset under increasing feature noise (single seed)

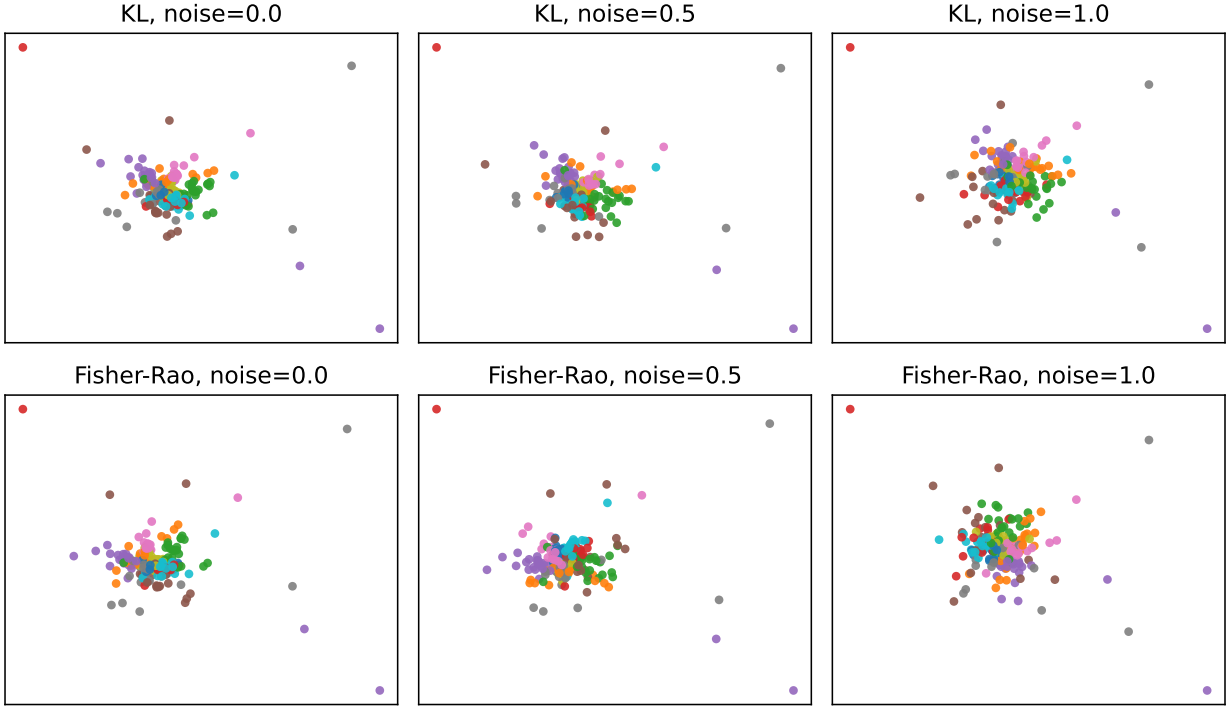


Figure 5: Qualitative t-SNE embeddings of `digits` (single seed) under three feature noise levels for the two objectives. The same initialization and optimizer are used; the two objectives produce visually similar embeddings.

once the geodesic angle reaches π .

Why are neighborhood metrics indistinguishable? KL and Fisher–Rao squared agree to leading order at $p = q$; they only differ globally. During optimization, especially in the late phase, Q approaches P for the high-probability pair indices, and the two objectives produce similar gradients there. The neighborhood metrics (trustworthiness, recall, kNN accuracy) measure agreement on these high-probability pairs, so they are close to insensitive to the global tail behavior that distinguishes the two objectives.

Why does KL especially dominate on Gaussian blobs at high noise? On well-separated isotropic clusters with strong noise, the clean affinity matrix P has a sparse block-diagonal structure: high mass within clusters, near-zero mass between clusters. Recovering this structure from a noisy $\tilde{X}$ requires the objective to push between-cluster Q_{ij} aggressively toward zero. KL’s unbounded penalty on missed-mass deviations does this naturally; Fisher–Rao’s bounded penalty does not push as hard. In high-dimensional, less cleanly clustered settings (`digits`, MNIST), there is less sharp block structure to recover, so this advantage attenuates.

When does boundedness help? The structured affinity-corruption experiment in Section 4.6 flips the sign of the story. There, the problem is not that the objective must recover a sharp

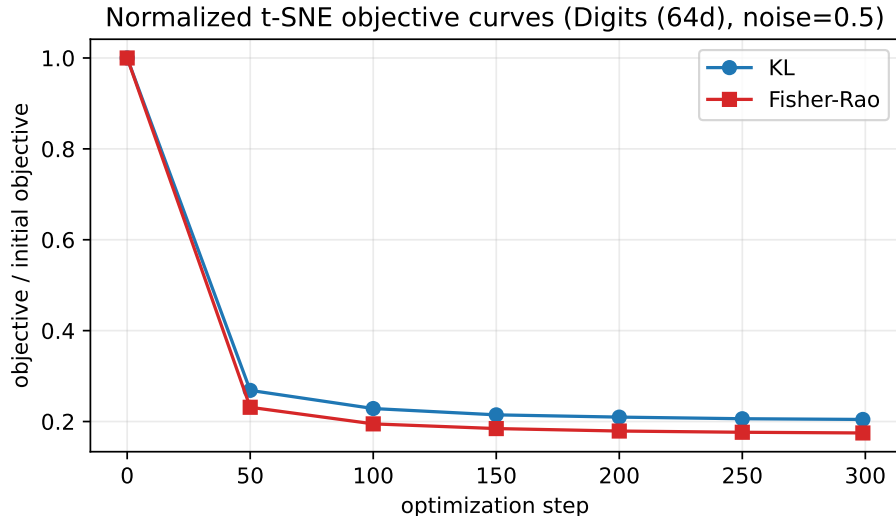


Figure 6: Normalized t-SNE objective curves on `digits` at noise $\alpha = 0.5$. Both objectives are well-behaved under Adam; the relative decrease rates differ because the underlying loss landscapes have different scales.

clean block structure from noisy features; it is that the target distribution itself contains false high-probability edges. KL’s unbounded pressure then becomes a liability because it faithfully pulls those false pairs together. Fisher–Rao is less obedient to extreme target mass and therefore better preserves the clean neighborhoods. This does not rescue Fisher–Rao as a default t-SNE objective, but it identifies a concrete use case: robust affinity matching when the affinity graph is contaminated by spurious high-confidence edges.

What did the VAE study add? The VAE results test a different hypothesis from t-SNE. For Gaussian posteriors, the categorical bounded-codomain argument does not apply; Fisher–Rao is unbounded on the Gaussian manifold. The relevant question is whether intrinsic Gaussian geometry gives a better regularization trade-off after beta tuning. In our compact fully connected VAE, the answer is again mostly negative: Fisher–Rao is competitive, but it does not reliably improve reconstruction, latent classification, posterior matching, generation, or robustness. Its practical value is diagnostic. It exposes how much of a claimed improvement comes from regularizer geometry versus beta scale, and it makes posterior-geometry trade-offs visible.

When could Fisher–Rao still matter? The current VAE experiment is small: two image datasets, five seeds, two epochs, and a fully connected architecture. Fisher–Rao may still be useful in settings where the Gaussian posterior geometry is more central: hierarchical VAEs, high-dimensional latents, annealed regularization schedules, or models where posterior variance dynamics are unstable under KL. The benchmark code now supports that follow-up directly through beta sweeps, stability tracking, corruption evaluation, and cached resumable runs.

Methodological takeaway. The wider methodological point is that information-geometric replacements for KL should be evaluated with paired multi-seed protocols and explicit significance tests rather than single runs. The infrastructure for doing so is small: our entire benchmark, including dataset loading, loss implementation, and statistical analysis, is < 500 lines of Python.

Noisy-affinity stress test means across seeds

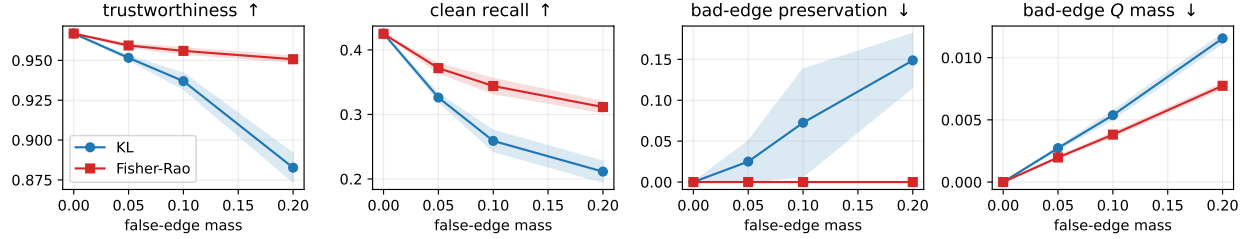


Figure 7: Noisy-affinity stress test under injected false high-probability edges. Unlike feature-noise corruption, this directly contaminates the target pair distribution. KL increasingly preserves the injected bad edges as false-edge mass grows, while Fisher–Rao maintains higher clean-neighborhood metrics and assigns less embedding probability to the corrupted pairs.

Structured false-affinity robustness: Fisher-Rao resists corrupted edges

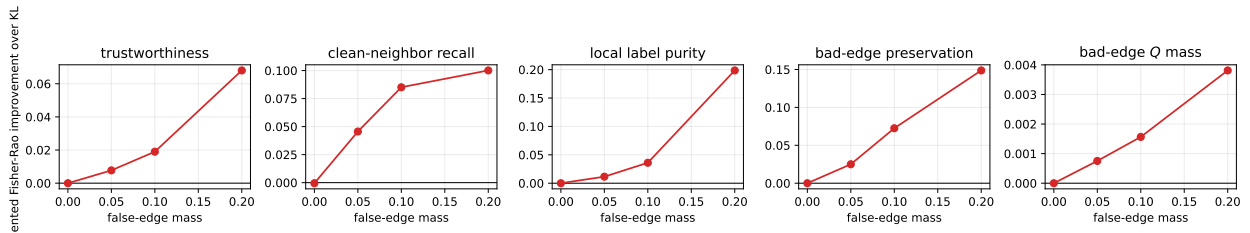


Figure 8: Oriented Fisher–Rao improvement over KL in the noisy-affinity stress test. The last two panels multiply lower-is-better metrics by -1 , so positive values consistently indicate a Fisher–Rao advantage.

7 Limitations

The compact 200-point t-SNE size means the affinity matrix is 200×200 , well below practical t-SNE deployments at 10^4 – 10^5 points. Although nothing in the gradient code prevents scaling, the empirical patterns may shift for larger neighborhoods. We use a fixed median-distance bandwidth instead of perplexity-tuned bandwidths; this matches the simple synthetic baseline but is not the canonical t-SNE preprocessing. The robustness corruption is isotropic Gaussian noise; structured corruption (occlusions, label noise, manifold warping) might give a different ordering. The structured affinity-corruption stress test is currently a pilot with five seeds; its effect sizes motivate a larger confirmation study with more seeds, datasets, and corruption mechanisms such as hub, block, and boundary false edges. Finally, we do not study early-exaggeration or annealing schedules, both of which interact non-trivially with objective scale. Each of these limitations defines a clean follow-up experiment in the same multi-seed framework.

8 Conclusion

The Fisher–Rao geodesic distance is a symmetric, triangle-inequality-respecting, parameterization-invariant, bounded-codomain alternative to KL divergence for comparing probability distributions. Despite this geometric appeal, when used as a t-SNE objective and evaluated with the same paired-comparison protocol that is now standard in empirical ML, it does not outperform KL: across 3

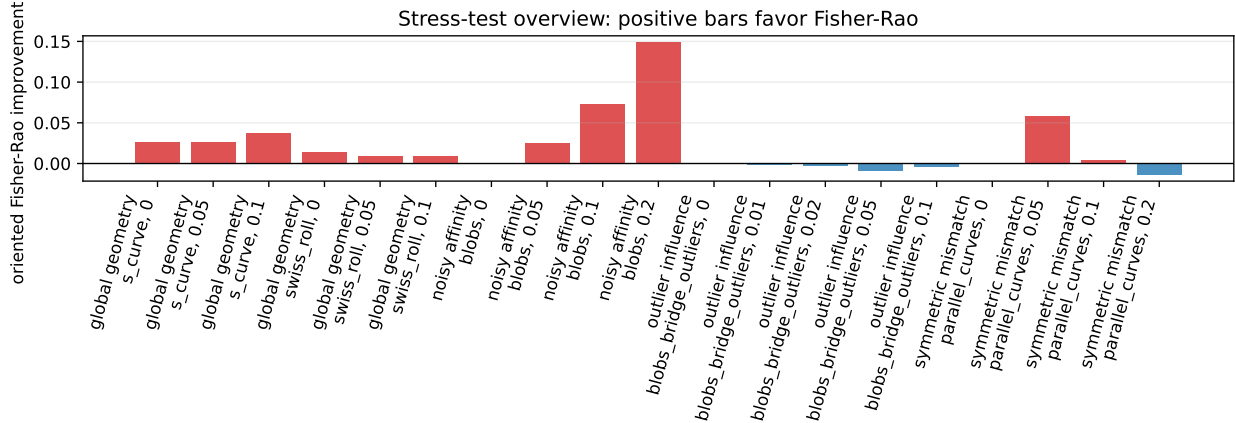


Figure 9: Stress-test overview. Each bar is oriented so positive means Fisher–Rao improves the most diagnostic metric for that stress family. The clearest signal is the noisy-affinity false-edge benchmark; other stress families are weaker or mixed.

datasets, 5 corruption levels, and 10 seeds, KL significantly wins 13 out of 60 metric-cells (mostly silhouette and the strong-noise regime on simple Gaussian clusters), Fisher–Rao significantly wins zero, and the remaining 44 are statistical ties. Practitioners should continue to use KL for clean t-SNE-style affinity matching. The new structured-corruption stress test refines this negative result: Fisher–Rao can help when the affinity target contains sparse false-neighbor edges, where KL’s unbounded pressure overfits spurious high-confidence mass. In VAEs, Fisher–Rao is competitive after separate beta tuning but does not produce a reliable improvement in the compact study reported here. Practitioners should therefore treat Fisher–Rao as an experimental regularizer and diagnostic geometry, not as a default replacement for KL.

A VAE Benchmark Details

The VAE benchmark is implemented in the dedicated VAE benchmark script and aggregated by the matching VAE aggregation script. It writes raw per-configuration metrics, mean/std beta summaries, selected best-beta rows, and paired tests under **reports/results**. The benchmark is resumable: completed (dataset, seed, regularizer, β) cells are skipped unless **--force** is passed.

Table 4: Best-beta VAE paired differences, Fisher–Rao minus KL. Positive BCE means Fisher–Rao reconstructs worse; positive latent kNN means Fisher–Rao has better latent classification.

Dataset	BCE/pix	Latent kNN	Posterior MMD	Noise BCE
MNIST	+0.0018	−0.0016	+0.0327	+0.0042
Fashion-MNIST	+0.0025	+0.0133	−0.0002	+0.0026

The table reports mean paired deltas across five seeds after beta selection. None of the listed effects reaches $p < 0.05$ in the two-sided Wilcoxon signed-rank test, so the evidence does not support a claim that Fisher–Rao improves the VAE objective in this setting.

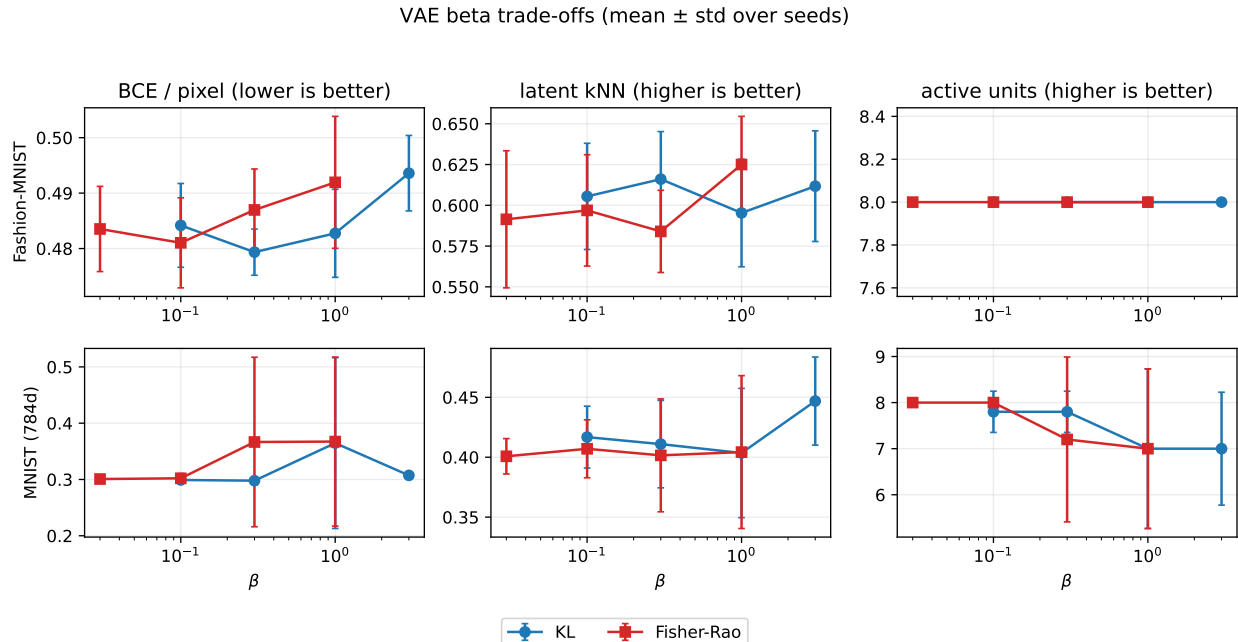


Figure 10: VAE beta trade-offs across reconstruction, latent kNN accuracy, and active units. Curves show mean ± 1 standard deviation across five seeds for each dataset and regularizer.

B Dataset and Corruption Details

B.1 Blobs

Generated with `sklearn.datasets.make_blobs(n_samples=200, n_features=8, centers=5, cluster_std=1.6, random_state=0)`, then z-scored. Five Gaussian centers in $\mathbb{R}^8$.

B.2 Digits

A 200-point random subsample (numpy default RNG seeded 0) from sklearn’s `load_digits` bundle, which contains 1797 images of 8×8 pixel handwritten digits, 10 classes, and 64 raw features. Z-scored to zero mean and unit variance.

B.3 MNIST

A 200-point random subsample (numpy default RNG seeded 0) from the MNIST training split, each image flattened to 784 raw pixel features and z-scored.

B.4 VAE datasets

The VAE benchmark uses torchvision MNIST and Fashion-MNIST with raw pixel intensities in $[0, 1]$. For each seed, 1024 training images and 512 test images are sampled without replacement. KMNIST support is implemented in the benchmark script, but the KMNIST download timed out during this run and is therefore not included in the reported VAE tables.

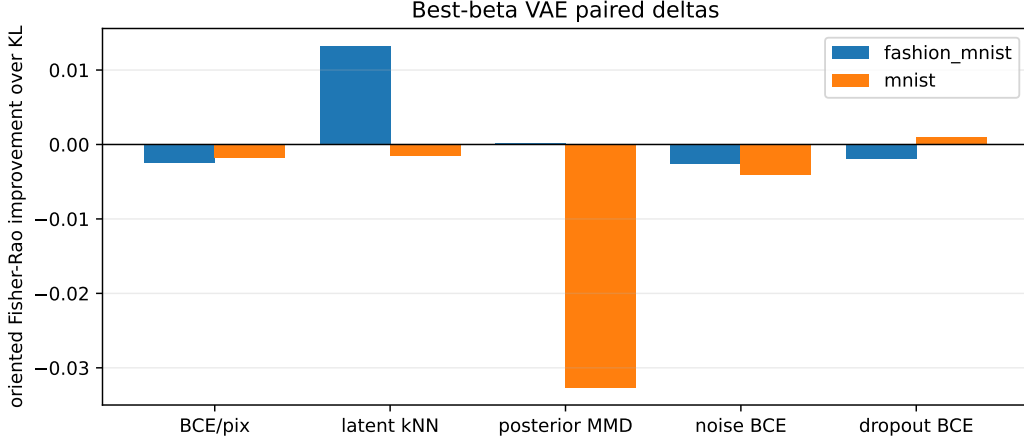


Figure 11: Best-beta paired VAE deltas. Values are oriented so positive means Fisher–Rao is better: lower BCE/MMD is multiplied by -1 , while higher latent accuracy is left positive.

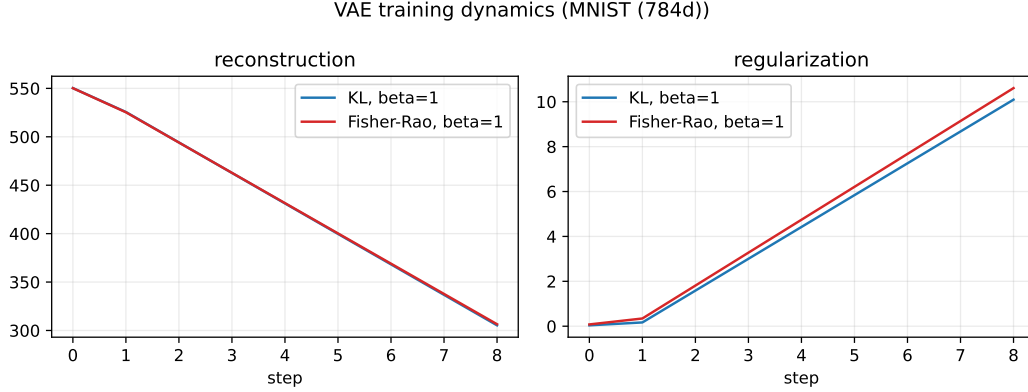


Figure 12: VAE training dynamics on MNIST for the representative $\beta = 1$ setting. Raw KL and Fisher–Rao regularization values are not directly comparable in scale, but both optimize smoothly under Adam after gradient clipping.

B.5 Robustness corruption

For each noise level $\alpha \in \{0.0, 0.25, 0.5, 0.75, 1.0\}$ and each seed $s \in \{101, 202, \dots, 1010\}$, we draw a fresh standard normal noise tensor of the same shape as the standardized feature matrix and form $\tilde{X} = X + \alpha \epsilon$. The seed for the noise tensor is $1000s + \lfloor 1000\alpha \rfloor$ to keep noise shocks reproducible and distinct across cells.

C Training Dynamics Details

All t-SNE benchmark runs use Adam at learning rate 0.05 for 300 steps, with gradient descent on the 200×2 embedding matrix. Optimizer state is reset for each (dataset, noise, seed, objective) run. The high-dimensional bandwidth is set per dataset to $\sigma = \text{median}_{i < j} \|x_i - x_j\| / \sqrt{2}$ on the clean standardized data, giving $\sigma \approx 2.81$ for blobs, $\sigma \approx 7.0$ for digits, and $\sigma \approx 21.8$ for MNIST.

The full benchmark, including aggregation and statistical analysis, runs in approximately two and

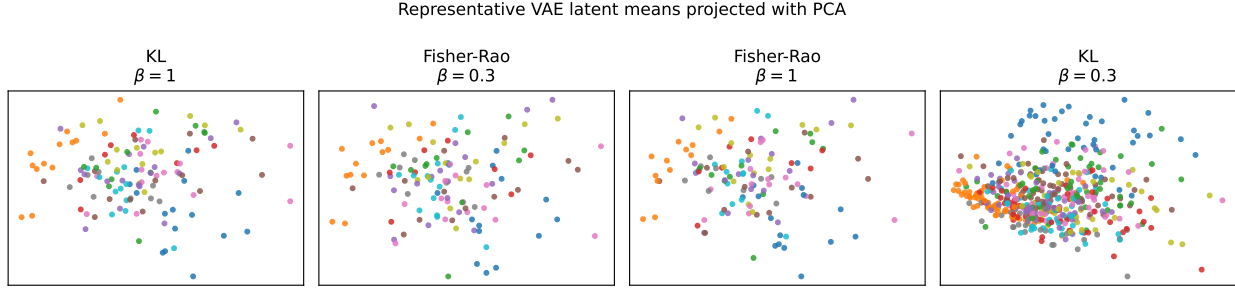


Figure 13: Representative VAE latent means projected to two dimensions with PCA. The plots visualize posterior geometry rather than serving as a quantitative metric; paired numerical metrics are reported in the CSV tables.

a half minutes on Apple Silicon MPS using the script `experiments/paper_benchmark.py` followed by `experiments/aggregate_results.py`, with figures regenerated by `reports/generate_figures.py`. End-to-end the paper can be rebuilt with `make paper-all`.

The VAE benchmark uses `experiments/vae_benchmark.py`. The full default CLI supports ten seeds and ten epochs; the reported local run used five seeds and two epochs to keep the experiment interactive on Apple Silicon while still providing paired beta-tuned comparisons. The command is recorded in the repository history and all output CSVs are stored under `reports/results`.

References

- [1] Shun-ichi Amari. *Information Geometry and Its Applications*. Springer, 2016.
- [2] Norman Cliff. Dominance statistics: Ordinal analyses to answer ordinal questions. *Psychological Bulletin*, 114(3):494–509, 1993.
- [3] S. Kullback and R. A. Leibler. On information and sufficiency. *The Annals of Mathematical Statistics*, 22(1):79–86, 1951.
- [4] Yann LeCun, Léon Bottou, Yoshua Bengio, and Patrick Haffner. Gradient-based learning applied to document recognition. *Proceedings of the IEEE*, 86(11):2278–2324, 1998.
- [5] Frank Nielsen. An elementary introduction to information geometry. *Entropy*, 22(10):1100, 2020.
- [6] Frank Nielsen. A simple approximation method for the Fisher–Rao distance between multivariate normal distributions. *Entropy*, 25(4):654, 2023.
- [7] C. R. Rao. Information and the accuracy attainable in the estimation of statistical parameters. *Bulletin of the Calcutta Mathematical Society*, 37:81–91, 1945.
- [8] Laurens van der Maaten and Geoffrey Hinton. Visualizing data using t-sne. *Journal of Machine Learning Research*, 9:2579–2605, 2008.
- [9] Jarkko Venna, Jaakko Peltonen, Kristian Nybo, Helena Aidos, and Samuel Kaski. Information retrieval perspective to nonlinear dimensionality reduction for data visualization. *Journal of Machine Learning Research*, 11:451–490, 2010.

- [10] Frank Wilcoxon. Individual comparisons by ranking methods. *Biometrics Bulletin*, 1(6):80–83, 1945.